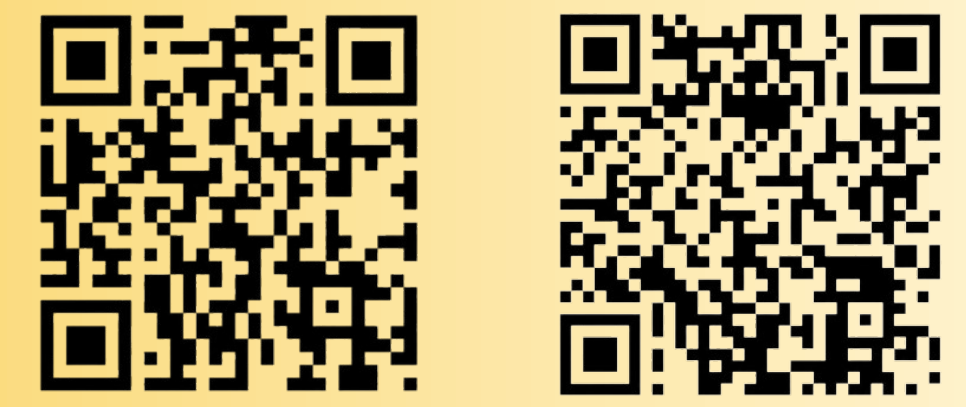


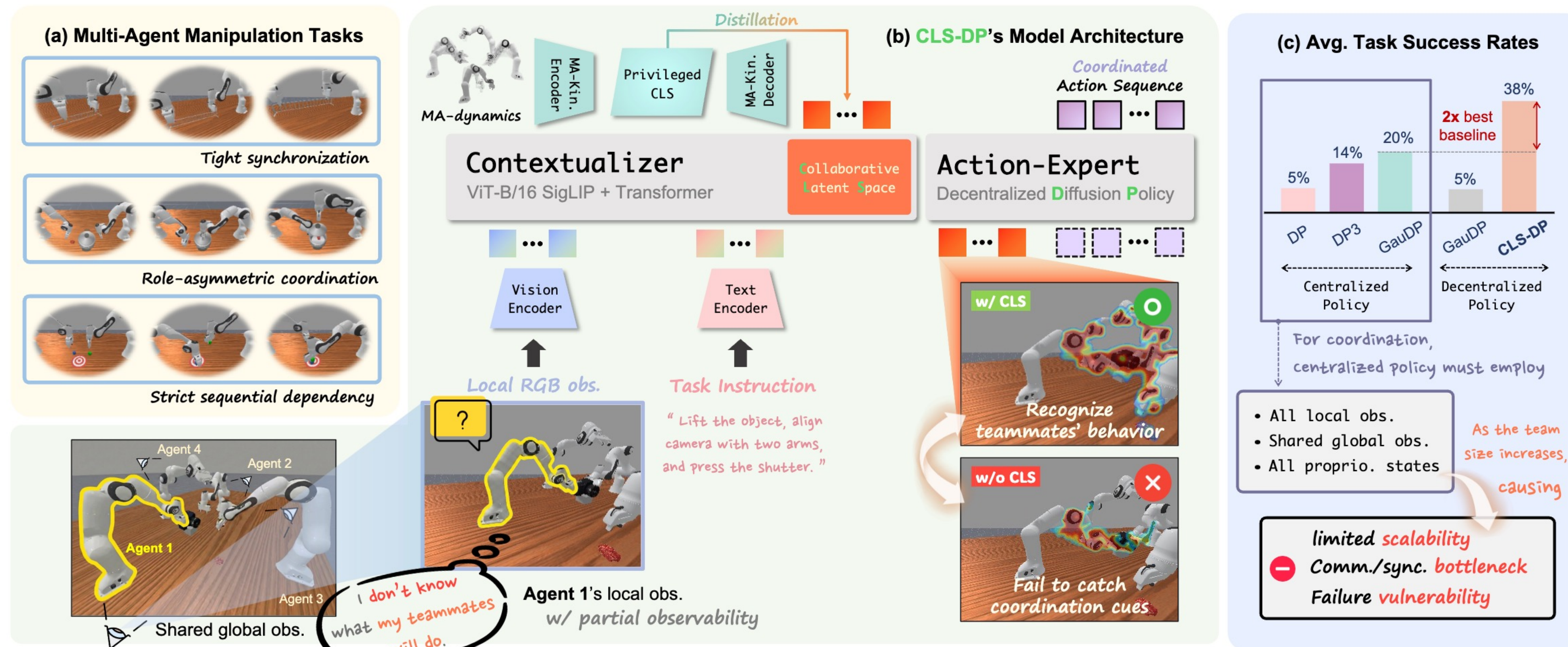
Distilling Collaborative Dynamics into Latent Space for Implicit Coordination in Decentralized Multi-Agent Manipulation

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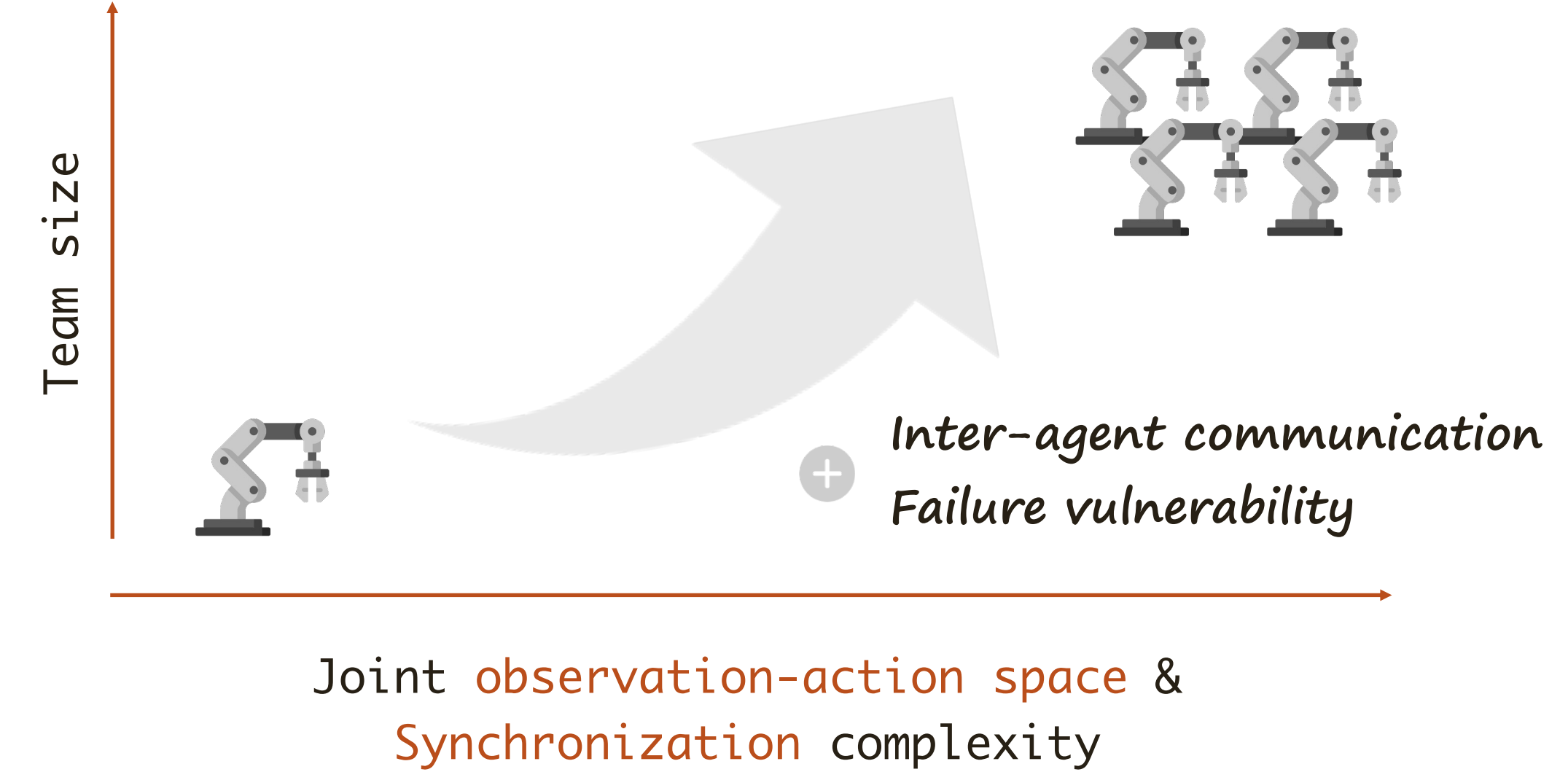


Project Page LinkedIn

1. Overview

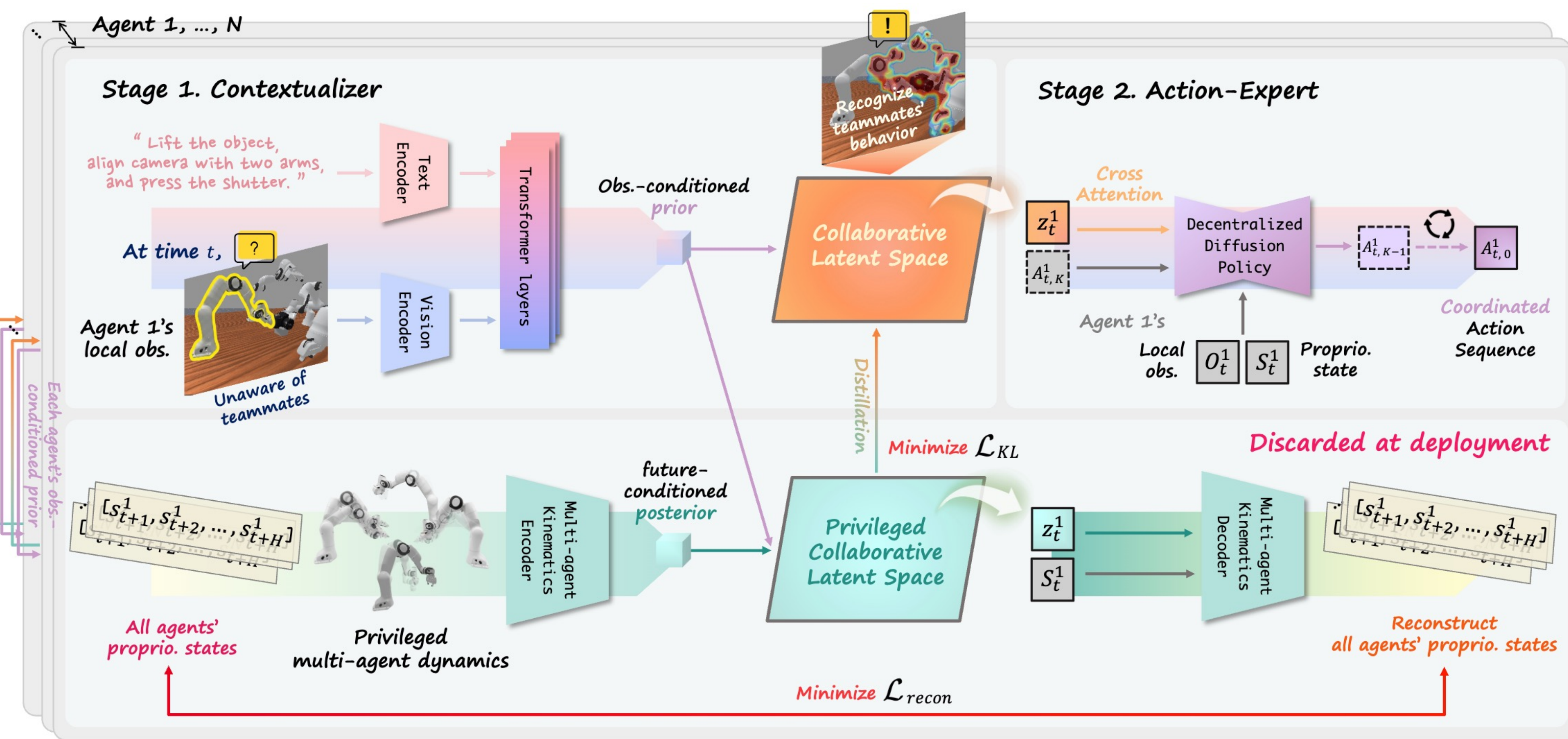


Coordinating multiple arms is hard:



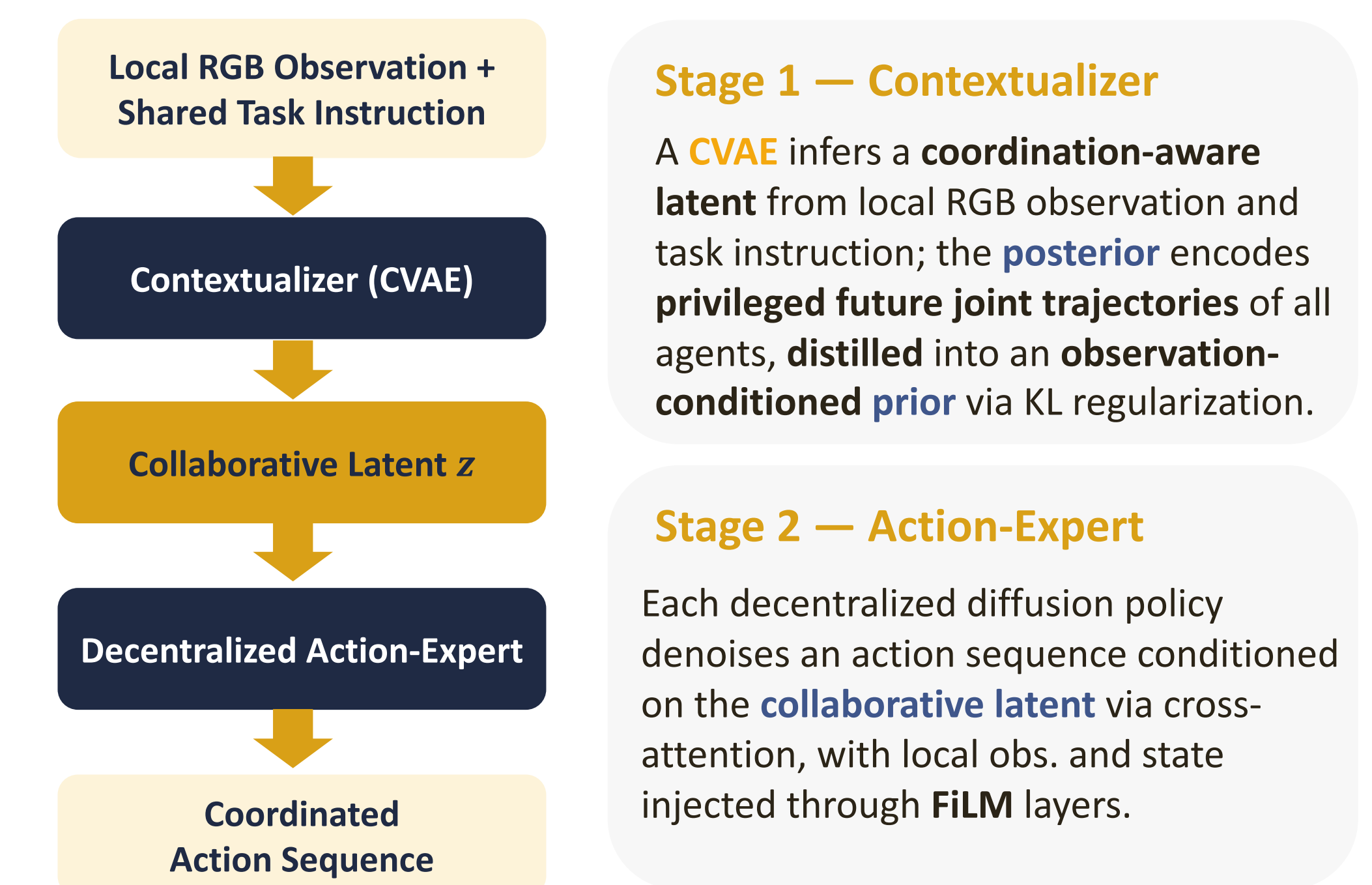
CLS-DP instead enables implicit coordination under partial observability — each agent infers a **collaborative latent** from its local RGB observation alone, with **no shared views, global state, or communication** required.

2. Method



Key Contributions

- Truly Decentralized**: At inference, each agent acts from only its local RGB view and a shared instruction — no global views, no explicit state, no communication.
- Scalable by Design**: Per-agent computational cost stays constant as team size grows, yielding the best success-rate-per-parameter across multiple agents.



3. Experiments

I. Quantitative Results — Success Rates & Model Efficiency

AVERAGE SUCCESS RATES ACROSS BENCHMARK TASKS OVER 100 UNSEEN EPISODES. THE RESULTS ARE RANKED AS 1ST, 2ND, 3RD.

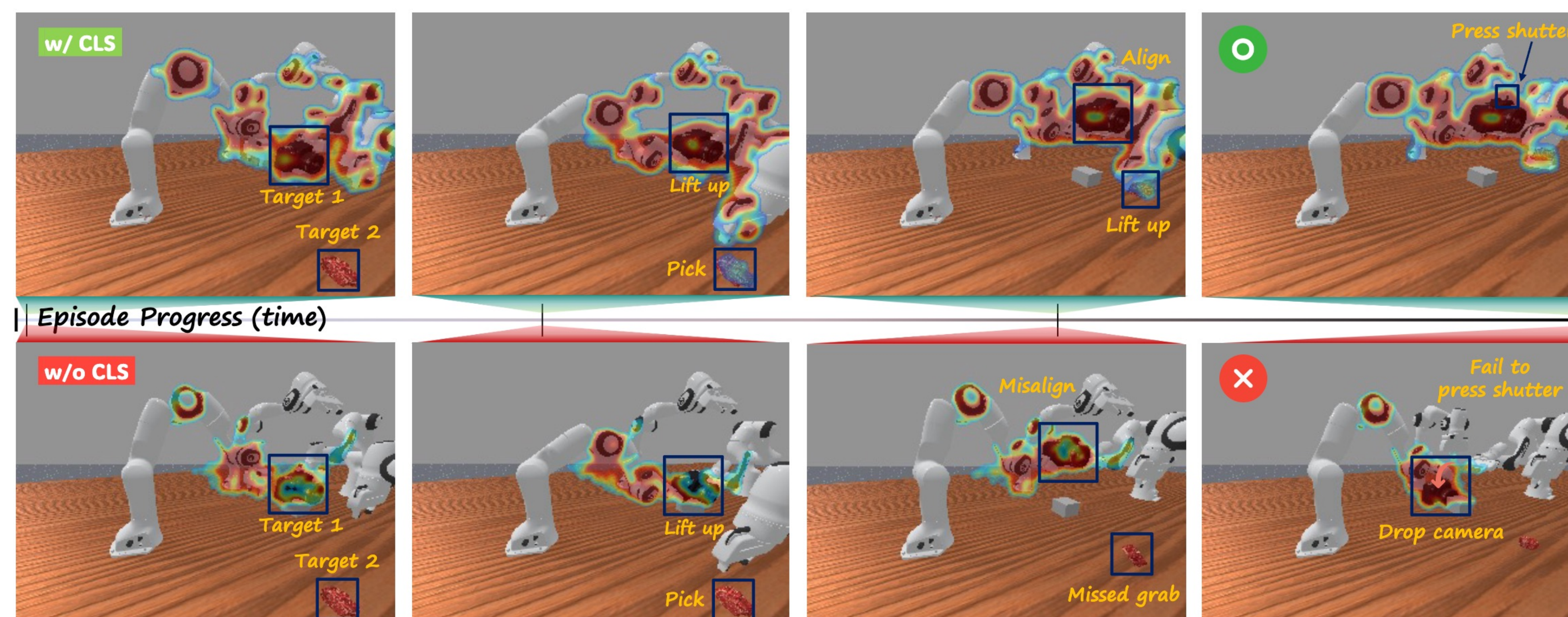
Methods	2 Agents			3 Agents		4 Agents	Total
	Lift Barrier	Place Food	Stack Cube	Camera Align.	Stack Cube	Take Photo	
Centralized Policies							
DP [7]	9%	12%	6%	3%	0%	0%	5%
LargeDP [7]	60%	12%	4%	29%	0%	0%	18%
2D Dense Policy [39]	3%	2%	0%	0%	0%	9%	2%
DP3 (XYZ) [40]	30%	21%	1%	3%	0%	9%	11%
DP3 (XYZ+RGB) [40]	31%	25%	1%	18%	0%	11%	14%
3D Dense Policy [39]	28%	18%	0%	0%	0%	7%	9%
Global GauDP [38]	72%	15%	2%	26%	0%	3%	20%
Decentralized Policies							
Local GauDP [38]	3%	12%	0%	15%	0%	2%	5%
CLS-DP (Ours)	61%	43%	39%	55%	20%	8%	38%
Ours w/o CLS	14%	5%	14%	7%	8%	3%	9%

EFFICIENCY COMPARISONS (↑):
AVERAGE SUCCESS RATE (↑) [%] PER PARAMETERS (↓) [M].

Method	2 Agents	3 Agents	4 Agents
DP [7]	0.0693	0.0092	0.0000
LargeDP [7]	0.0351	0.0152	0.0000
GauDP-G [38]	0.0395	0.0166	0.0037
GauDP [38]	0.0068	0.0035	0.0009
CLS-DP+K (ours)	0.2013	0.1124	0.0186
CLS-DP (ours)	0.2056	0.1148	0.0190

CLS-DP nearly doubles the best centralized baseline — fully decentralized and parameter-efficient.

II. Qualitative Findings — Coordination Failures Resolved



01 Object Drop Resolved

Without the collaborative latent, unsynchronized lifting causes one agent to drop the object. CLS-DP tightly synchronizes both agents' lift timing.

02 Balanced Lifting

Imbalanced asymmetric lifting causes a backward tilt and drop. CLS-DP coordinates lift force across agents to keep the object level.

03 Collision Avoided

Violating strict sequential dependency causes inter-agent collisions during simultaneous picking. CLS-DP respects the required pick order.